Siddesh Voleti

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OBJECTIVE

Electrical Engineering undergraduate seeking an internship in analog/digital electronics, embedded systems, power systems, communications, or hardware design, with hands-on experience in PCB design, microcontrollers, power distribution, robotics, and system integration.

EDUCATION

B.S. in Electrical Engineering | Expected 2028
Iowa State University | Ames, Iowa
Electrical & Communication Engineering / ECE | 2 Years Completed | GPA: 7.16/10
Birla Institute of Technology and Science, Pilani

LEADERSHIP

Joint Secretary — Andhra Samithi | Jan 2026
Birla Institute of Technology and Science, Pilani
• Coordinated student activities and worked across teams to support planning, execution, and organizational operations.

PROJECTS

VELOS — 15 kg Drum-Spinner Combat Robot | Electronics Lead & System Integration | April 2026
• Led the complete electrical subsystem through a 10-day concept-to-build cycle, integrating drivetrain, weapon, RC controls, batteries, ESCs, and embedded electronics.
• Designed and validated power distribution, routing battery power to drive/weapon ESCs and regulated power to control electronics; integrated a hardwired master kill switch for emergency isolation.
• Designed a KiCad PCB for ESP32-based control and signal distribution, integrating RC inputs, sensor interfaces, ESC signals, and power monitoring.
• Developed ESP32 embedded control for RC command processing, ESC control, sensor interfacing, system monitoring, communication-loss failsafe, and real-time operation.
• Fabricated a custom Y-splitter and wiring harnesses to simplify signal distribution and improve serviceability; performed soldering, integration, testing, and troubleshooting.
• Performed post-competition ESC overcurrent failure analysis, including investigation of PCB microfractures.

Color-Sensing Robotic Arm | January 2026
• Designed a 2-DOF RR servo-actuated robotic arm with coordinated PWM motion control for autonomous pick-and-place sorting.
• Engineered and calibrated an RGB LED-LDR color-sensing module with threshold-based detection for varying ambient-light conditions.
• Integrated Arduino control, servo actuators, electromagnetic gripper, and color sensing for autonomous object sorting.

DIY ESP32 Wireless Drawing Tablet | December 2025
• Built an ESP32-based wireless drawing tablet integrating touch/input sensing, coordinate processing, embedded firmware, and host-device communication.
• Programmed real-time input acquisition, coordinate mapping, and serial/wireless data transmission between the microcontroller and host device.
• Designed, assembled, debugged, and tested the electronic system using Embedded C, sensor interfacing, and real-time processing.

TECHNICAL SKILLS

Programming | C, C++, Python,Assembly
Microcontrollers | ESP32, Arduino, PWM, UART, I²C, SPI
Electronics | Analog & Digital Electronics, Power Electronics, Sensors, Motors, ESCs, Power Distribution
PCB & Hardware | KiCad, PCB Design, Circuit Debugging, Soldering, Wiring Harnesses, System Integration, Hardware Testing
Engineering Tools | LTspice, Cadence, AutoCAD
Fabrication | CNC Machining, 3D Printing, Robotics Fabrication
Coursework | Analog Electronics, Digital Design, Electronic Devices, Signals & Systems, Control Systems, Microelectronics, Electrical Machines, Electromagnetics & Microwave Engineering